ENVS 193DS Final

Steven Le

2026-06-04

Table of contents

GITHUB REPOSITORY LINK HERE

Homework set up

```
#read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggplot2)
library(ggeffects)
library(flextable)
library(here)
library(MuMIn)
library(DHARMa)
library(rstatix)
#stored nest data as an object called nests
nests <- read_csv(here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression, which is a generalized linear model with a binomial error distribution. In part 2, they used a chi-square test, which is used to identify if there is a relationship between two categorical variables.

b. Suggestions for rewriting

Part 1.

The distance to water edge does seem to predict the probability of American Avocet microhabitat use.

Part 2.

There does seem to be a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is the type of behavior they were displaying, which is a categorical variable. For the American Avocets, the researchers put behavior into four main categories: feeding (searching, probing, and dunking bills into the water), roosting (standing, sitting, preening), breeding (nest-building, incubating, brooding chicks, alarm calling, breeding display, copulation, or predator distraction display), and other (walking, swimming, or flushing). This variable might influence American Avocet habitat use as different behaviors can change the duration of how long they remain in a particular habitat.

Problem 2. Data analysis

a. Clean and wrangle

```
#created new object from nests
nests_clean <- nests |>
#cleaned column names
  clean_names() |>
#selected for tree height, ant_species, and case_control columns
  select(case_control, ant_species, height_cm) |>
#reordered names in ant_species column
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
#filtered ant_species column to only have three required species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
#created new column with full scientific name for each ant species
  mutate(scientific_name = case_match(ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
```

```r
    "BBR" ~ "Crematogaster nigriceps")) |>
#reordered full scientific name of ant species
  mutate(scientific_name = factor(scientific_name,
                                  levels = c("Crematogaster sjostedti",
                                             "Crematogaster mimosae",
                                             "Crematogaster nigriceps")))
#displayed structure of nests_clean data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 5 levels "AB","RRB","BBR",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```r
#displayed 10 random rows from the data frame but made it separate from all
↪  codes above by using separate pipe operator
nests_clean |>  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   case_control ant_species height_cm scientific_name
          <dbl> <fct>           <dbl> <fct>
 1            0 RRB              168. Crematogaster mimosae
 2            0 RRB              114. Crematogaster mimosae
 3            0 RRB              446  Crematogaster mimosae
 4            1 RRB              392. Crematogaster mimosae
 5            0 RRB              216  Crematogaster mimosae
 6            0 RRB              134. Crematogaster mimosae
 7            0 RRB              147  Crematogaster mimosae
 8            0 RRB              360. Crematogaster mimosae
 9            1 BBR              318. Crematogaster nigriceps
10            1 RRB              396  Crematogaster mimosae
```

b. Response variable

For the case_control column, the 1s mean that a tree does contain a nest and the 0s mean that a tree does not contain a nest (is an “available” tree for comparison).

c. Purpose of study

Tree height is a continuous (numeric) variable and could influence the probability of nest occurrence as if tree height were to increase, the probability of nest occurrence would increase because there would most likely be less predators higher up on trees (Results section, paragraph 2). Acacia ant species is a categorical (nominal variable) and could influence the probability of nest occurrence as if a particular ant species at a nesting site was more aggressive, the probability of nest occurrence could increase as it would reduce nest predation risks (Discussion section, paragraph 1).

d. Table of models

Model number	Tree height	Ant species	Model description
0	X	X	All predictors (no interaction term, +)
1	X	X	All predictors (interaction term, *)

e. Run the models

```
#model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ height_cm + scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)

#model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ height_cm * scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)
```

f. Select the best model

```
#calculated AIC for each model
AICc(
  nest_model0,
  nest_model1) |>
#arranged output in order of AIC
```

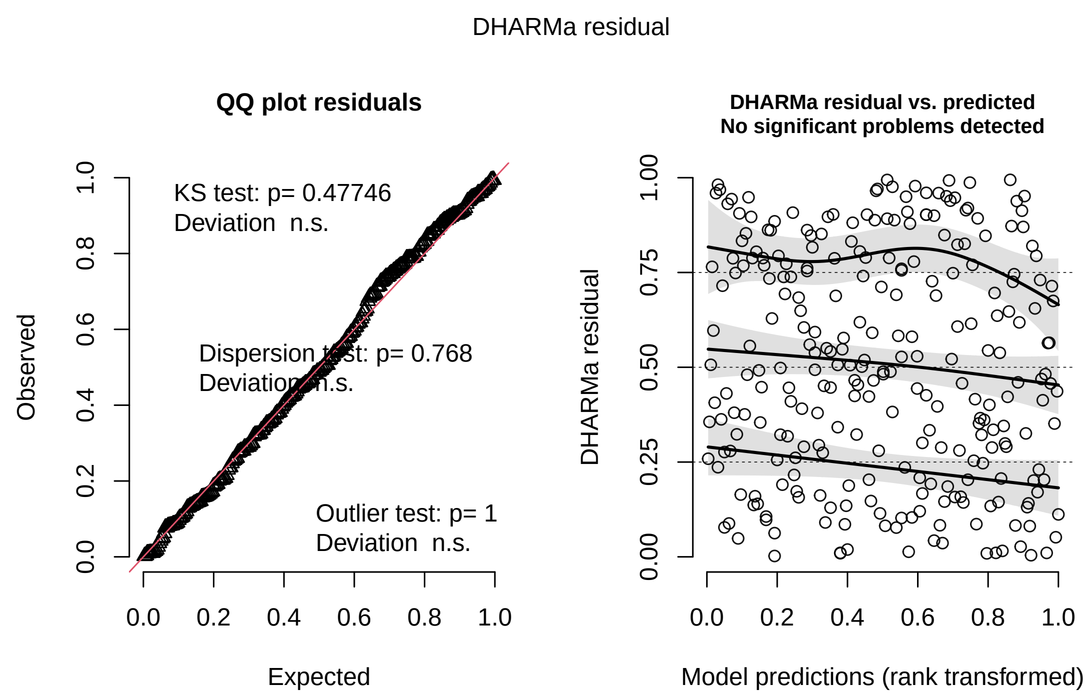
```
  arrange(AICc)
```

```
             df     AICc
nest_model1   6 238.1236
nest_model0   4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) was the model where tree height (cm) and acacia ant species were interacting terms for the probability of nest occurrence.

g. Check the diagnostics

```
#checked diagnostics for model 1 using DHARMa residuals
plot(simulateResiduals(nest_model1))
```



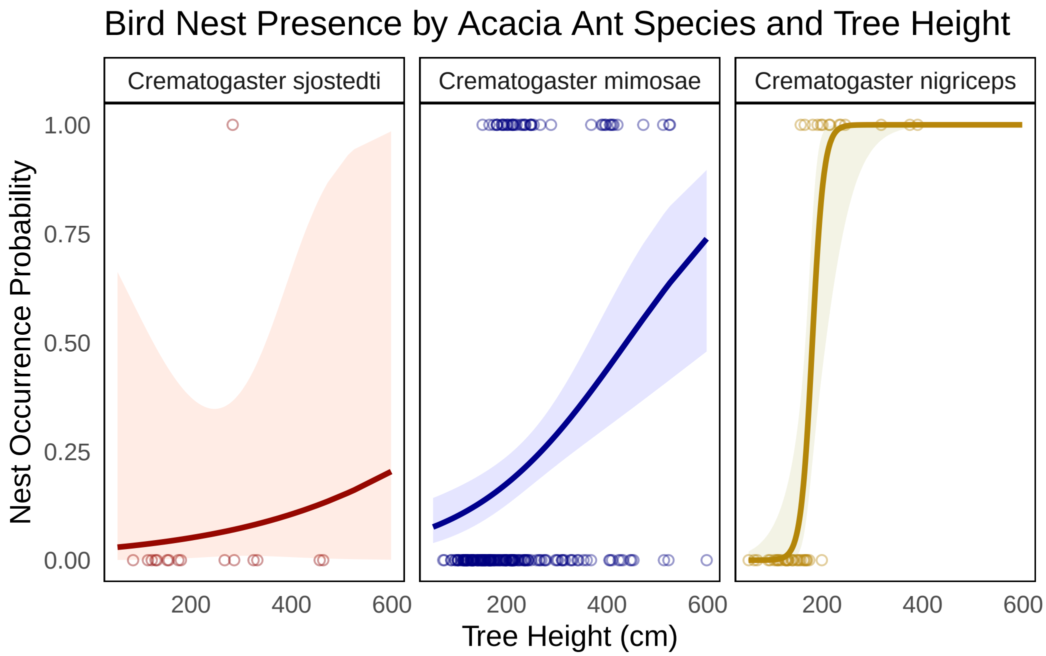
Most of the points on the QQ plot follow the reference line, so we can assume the data is roughly normally distributed. We can only believe the observations collected were independent and the response and we already know the response variable will be distributed following a binomial distribution (0 or 1). For the plot on the right, since it says that there are no significant problems detected, we can also assume that there is a linear relationship between transformed expected response (nest occurrence probability) in terms of link function and explanatory variables (tree height and ant species). Therefore, all of the assumptions for a generalized linear model (GLM) are met.

h. Visualize the model predictions

```
#created new object called preds
preds <- ggpredict(nest_model1,
#predictors
                   terms = c("height_cm[all]",
                             "scientific_name"))
#added scientific_name column for visualization
preds$scientific_name <- preds$group

#base layer: ggplot
ggplot(data = nests_clean,
       mapping = aes(x = height_cm,
                     y = case_control,
                     color = as.factor(scientific_name))) +
#relabelled x- and y-axes
  labs(x = "Tree Height (cm)",
       y = "Nest Occurrence Probability",
#colored by ant species
       color = "scientific_name",
#added title to figure
       title = "Bird Nest Presence by Acacia Ant Species and Tree Height") +
#changed colors for each ant species plot
  scale_color_manual(values = c("Crematogaster sjostedti" = "darkred",
                                "Crematogaster mimosae" = "navyblue",
                                "Crematogaster nigriceps" = "darkgoldenrod"))
                                 ↪  +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "orangered",
                                "Crematogaster mimosae" = "blue",
                                "Crematogaster nigriceps" = "yellow4")) +
#used a different theme from default
  theme_minimal() +
#removed background
theme(panel.background = element_blank(),
```

```
#removed major gridlines
        panel.grid.major = element_blank(),
#removed minor gridlines
        panel.grid.minor = element_blank(),
#removed legend
        legend.position = "none",
#added borders to each panel
        panel.border = element_rect(color = "black",
                                    fill = NA,
                                    linewidth = 0.5),
#added borders to each ant species name title
        strip.background = element_rect(color = "black",
                                        fill = "white",
                                        linewidth = 0.5)) +
#first layer: geom point
  geom_point(mapping = aes(x = height_cm, #x-axis: tree height
                           y = case_control, #y-axis: nest occurrence
                           color = scientific_name), #set to selected colors
#changed shape
             shape = 21,
#made slightly transparent
             alpha = 0.4,
#ignored global aes for colors
             inherit.aes = FALSE) +
#second layer: geom line
  geom_line(data = preds, #data from model 1 prediction
            mapping = aes(x = x, #x-axis: tree height
                          y = predicted, #y-axis: nest probability
                          color = group), #colored by ant species
            linewidth = 1) +
#third layer: CI ribbon
  geom_ribbon(data = preds, #data from model 1 prediction
              mapping = aes(x = x, #x-axis: tree height
                            ymin = conf.low, #set low value
                            ymax = conf.high, #set high value
                            fill = group), #filled by ant species
              inherit.aes = FALSE, #ignored global aes
              alpha = 0.1) + #made transparent
#made new panel for each ant species
  facet_wrap(~ scientific_name)
```


Bird Nest Presence by Acacia Ant Species and Tree Height
Crematogaster sjostedti
Crematogaster mimosae
Crematogaster nigriceps
Nest Occurrence Probability
1.00
0.75
0.50
0.25
0.00
200
400
600
Tree Height (cm)

i. Write a caption for your figure

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

b. Designing an analysis

c. Check any assumptions and run your analysis

d. Create a visualization

e. Write a caption

f. Write about your results

g. Sharing your affective visualization

COMPLETED